



MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust®, Mangalore)

Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE, New Delhi

Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course Title: Computational Thinking & Programming

Course Code: 23ESCC111

List of Lab Programs

1. C program to read a character in upper case and then print it in lower case.
2. Vibha has a piggy bank where she regularly saves some coins. Assume the denominations of Notes as: 10,20,50,100 There are various numbers of Notes of each denomination. Input denominations for these Notes and calculate the total amount of money she saved in Rupees.
3. C program to find whether the given year is a leap year.
4. C program to read the coefficients for the equation $ax^2+bx+c=0$ and display the roots of the quadratic equation with appropriate messages.
5. C program to find if the given number is Palindrome or not.
6. C program to evaluate the Taylor series $f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{7}{7!} + \dots$, to compute $\sin(x)$.
7. C Program to read two 1-dimensional arrays, add the elements, and store the result in the third 1-dimensional array.
8. C Program to search a given element in an array using the Binary Search technique. Assume that the array is already sorted in an ascending order.
9. C Program to sort n elements in an array using Bubble Sort.
10. C program to check whether the given string is Palindrome or not
11. C Program to perform any 4 string manipulation functions using switch.
12. Write a C Program to swap two numbers using call by value and call by reference
13. Develop a program using dynamic memory to compute: the sum, mean, and standard deviation of all elements for n real numbers.
14. C program to introduce 2D Array manipulation. Implement Matrix multiplication and ensure the rules of multiplication are checked by using functions.
15. Implement structures to read, write, and compute average- marks of the students, list the students scoring above average marks for a class of N students.
16. Write a C program to copy a text file to another, read both the input file name and target file name



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Develop a C program to read a character in upper case and then print it in lower case.

```
#include<stdio.h>
```

```
void main()
```

```
{
```

```
    char upper,lower;
```

```
    printf("\n Enter any character in uppercase:\n ");
```

```
    scanf("%c",&upper);
```

```
    lower=upper+32;
```

```
    printf("\n The character in lower case is: %c",lower);
```

```
}
```

Vibha has a piggybank in which she regularly saves the remaining amount from her expenditure. The denominations which she saves is of 10,20,50,100. Input the denominations for the same and calculate the total amount of money she saved.

```
#include<stdio.h>

void main()
{
    int ten, twenty, fifty,hundred,total_amt=0;
    printf("\n Enter the number of denominations for Rs 10: ");
    scanf("%d",&ten);
    printf("\n Enter the number of denominations for Rs 20: ");
    scanf("%d",&twenty);
    printf("\n Enter the number of denominations for Rs 50: ");
    scanf("%d",&fifty);
    printf("\n Enter the number of denominations for Rs 100: ");
    scanf("%d",&hundred);
    total_amt= ten*10+ twenty*20+ fifty*50+hundred*100;
    printf("\n Total amount in the piggybank= %d rupees",total_amt);
}
```



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C program to find whether the given year is leap year or not check end of centuries.

```
#include <stdio.h>

void main()
{
    int year;
    printf("\n Please Enter the year: ");
    scanf("%d", &year);
    if(((year % 4 == 0) && (year % 100 !=0)) || (year % 400==0))
        printf("\n %d is a Leap Year. \n", year);
    else
        printf("\n %d is not a Leap Year. \n", year);
}
```

Develop a C program to read the coefficients for the equation $ax^2+bx+c=0$ and display the roots of the quadratic equation with appropriate messages.

Program:

```
#include<stdio.h>
#include<stdlib.h>
#include<math.h>
void main()
{
    float a,b,c,x1,x2,disc;
    printf("Enter the values of a,b,c\n");
    scanf("%f%f%f",&a,&b,&c);
    if(a==0)
    {
        printf("Invalid Input\n");
        exit(0);
    }
    disc=b*b-4*a*c;
    if(disc>0)
    {
        printf("Roots are Real and Distinct\n");
        x1=(-b)+sqrt(disc)/(2*a);
        x2=(-b)-sqrt(disc)/(2*a);
        printf("Root1= %f\n Root2= %f\n",x1,x2);
    }
    else if(disc==0)
    {
        printf("Roots are Real and Equal\n ");
        x1=(-b)/(2*a);
        printf("Root1=Root2=%f\n",x1);
    }
    else
    {
        printf("Roots are Imaginary\n");
        x1=(-b)/(2*a);
        x2=(sqrt(fabs(disc)))/(2*a);
        printf("Root1= %f +i %f\n",x1,x2);
        printf("Root2= %f -i %f\n",x1,x2);
    }
}
```

Develop a program to find the reverse of a positive integer and check for palindrome or not. Display appropriate messages

```
#include<stdio.h>
void main()
{
    int num,temp,rev,rem;
    printf("Enter an integer number\n");
    scanf("%d",&num);
    rev=0;
    temp=num;
    while(num!=0)
    {
        rem=num%10;
        num=num/10;
        rev=((rev*10)+rem);
    }
    printf("%d after reversing is %d\n",temp,rev);
    if(temp==rev)
    {
        printf("%d is a palindrome\n",temp);
    }
    else
    {
        printf("%d is not a palindrome\n",temp);
    }
}
```

Develop a C program to evaluate Taylor series $f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$, to compute $\sin(x)$.

```
#include<stdio.h>
#include<math.h>
#define pi 3.142
void main()
{
    int degree,i;
    float x,num,den,term,sum=0;
    printf("Enter the value of degrees\n");
    scanf("%d",&degree);
    x=(degree*pi)/180;
    num=x;
    den=1;
    i=2;
    do
    {
        term=num/den;
        sum=sum+term;
        num=-num*x*x;
        den=den*i*(i+1);
        i=i+2;
    }while(fabs(term)>0.000001);
    printf("sin(%d)= %f without using library function\n",degree,sum);
    printf("sin(%d)= %f using library function\n",degree,sin(x));
}
```

. Develop a C Program to read two 1-Dimensional arrays perform the addition of the same and store the resultant in the third array.

```
#include<stdio.h>

void main()
{
    int a[50],b[50],c[50], n,i;
    printf("Enter the number of elements\n");
    scanf("%d",&n);
    printf("Enter the elements for Array A\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the elements for Array B\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&b[i]);
    }
    printf("\n** ONE DIMENSIONAL ARRAY ADDITION **\n");
    for(i=0;i<n;i++)
    {
        c[i]=a[i]+b[i];
    }
    printf("\nThe resultant array is\n");
    for(i=0;i<n;i++)
    {
        printf("%d\t",c[i]);
    }
}
```


Develop a C Program to search a given element in an array using Binary Search technique.

```
#include<stdio.h>
void main()
{
    int a[100],n,i,low,high,mid,key,flag=0;
    printf("Enter the size of the array\n");
    scanf("%d",&n);
    printf("Enter %d elements in ascending order\n",n);
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the element to be searched\n" );
    scanf("%d",&key);
    low=0;
    high=n-1;
    while(low<=high)
    {
        mid=(low+high)/2;
        if(key==a[mid])
        {
            flag=1;
            break;
        }
        else
            if(key>a[mid])
                low=mid+1;
            else
                high=mid-1;
    }
    if(flag==1)
        printf("Element found at position %d\n",mid+1);
    else
        printf("Element not found\n");
}
```

Develop a C Program to sort n elements in an array using Bubble Sort.

```
#include<stdio.h>
void main()
{
    int a[50],n,i,j,temp;
    printf("Enter the number of elements\n");
    scanf("%d",&n);
    printf("Enter %d elements\n",n);
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("The entered elements are\n");
    for(i=0;i<n;i++)
    {
        printf("%d\t",a[i]);
    }
    printf("\n***** SORTING *****\n");
    for(i=1;i<n;i++)
    {
        for(j=0;j<n-i;j++)
        {
            if(a[j]>a[j+1])
            {
                temp=a[j];
                a[j]=a[j+1];
                a[j+1]=temp;
            }
        }
    }
    printf("The sorted elements are\n");
    for(i=0;i<n;i++)
    {
        printf("%d\t",a[i]);
    }
}
```

Develop a C program to check the given string is Palindrome or not

```
#include<stdio.h>
#include<string.h>
void main()
{
    char str1[50],str2[50];
    int i,j,len,x;
    printf("Enter a String\n");
    scanf("%s",str1);
    j=0;
    len=strlen(str1);
    for(i=len-1;i>=0;i--)
    {
        str2[j]=str1[i];
        j++;
    }
    str2[j]='\0';
    printf("The original String =%s\n",str1);
    printf("The reversed String= %s\n",str2);
    x=strcmp(str1,str2);
    if(x==0)
        printf("%s is a palindrome\n",str1);
    else
        printf("%s is not a palindrome\n",str1);
}
```

Develop a C Program to perform any 4 string manipulation functions using switch.

```
#include<stdio.h>
#include<string.h>
void main()
{
    char str1[20],str2[20];
    int k,d,choice;
    printf("\n***MENU***\n");
    printf("1. String Length\t 2. String copy\n");
    printf("3. String Concatenate\t 4. String Compare\n");
    printf("Enter your choice\n");
    scanf("%d",&choice);
    switch(choice)
    {
        case 1:    printf("\n**STRING LENGTH**\n");
                   printf("Enter a string:");
                   scanf("%s",str1);
                   k=strlen(str1);
                   printf("\nLength of the string=%d\n",k);
                   break;
        case 2:    printf("\n**STRING COPY**\n");
                   printf("Enter a string:");
                   scanf("%s",str1);
                   printf("The original string=%s\n",str1);
                   strcpy(str2,str1);
                   printf("the copied string=%s\n",str2);
                   break;
        case 3:    printf("\n**STRING CONCATENATE**\n");
                   printf("Enter the first string:");
                   scanf("%s",str1);
                   printf("Enter the second string:");
                   scanf("%s",str2);
```

```
        strcat(str1,str2);
        printf("The Concatenated string=%s\n",str1);
        break;
case 4:    printf("\n**STRING COMPARE**\n");
           printf("Enter the first string:");
           scanf("%s",str1);
           printf("Enter the second string:");
           scanf("%s",str2);
           k=strcmp(str1,str2);
           if(k==0)
               printf("Strings are same\n");
           else
               printf("Strings are different\n");
           break;
default:   printf("\n You have entered a wrong choice\n");
           break;
    }
}
```

Develop a C Program to swap two numbers using call by value and call by reference

```
#include <stdio.h>
void swap_val(int x, int y)
{
    int temp;
    temp = x;
    x = y;
    y = temp;
    printf("Values after swapping using call by value\n x = %d\t, y = %d\n",
x,y);
}

void swap_ref(int *x, int *y)
{
    int temp;
    temp = *x;
    *x = *y;
    *y = temp;
}

void main()
{
    int x,y;
    printf("Enter values for x and y\n");
    scanf("%d%d",&x,&y);
    printf("Values before swapping are\n x = %d\t,y = %d\n", x,y);
    printf("\n Demonstrating call by vale\n");
    swap_val(x,y);
    printf("\n Demonstrating call by reference\n");
    swap_ref(&x,&y);
    printf("Values after swapping using call by reference\n x = %d\t, y =
%d\n", x,y);
}
```

Develop a C Program using dynamic memory to compute the sum, mean and standard deviation of all elements for n real numbers.

```
#include<stdio.h>
#include<stdlib.h>
#include<math.h>
void main()
{
    int i,n;
    float mean,std,sum=0;
    float *ptr;
    printf("Enter the number of elements\n");
    scanf("%d",&n);
    ptr=(float*)malloc(n* sizeof(float));
    printf("Enter the numbers\n");
    for(i=0;i<n;i++)
    {
        scanf("%f",ptr+i);
        sum=sum+ptr[i];
    }
    printf("Sum= %f\n",sum);
    mean=sum/n;
    printf("Mean= %f\n",mean);
    sum=0;
    for(i=0;i<n;i++)
    {
        ptr[i]=mean-ptr[i];
        ptr[i]=ptr[i]*ptr[i];
        sum=sum+ptr[i];
    }
    sum=sum/n;
    std=sqrt(sum);
    printf("Standard Deviation= %f\n",std);
}
```

Develop a C Program to introduce 2D Array manipulation and implement Matrix multiplication and ensure the rules of multiplication are checked by using functions.

```
#include<stdio.h>
int a[50][50],b[50][50],c[50][50],m,n,p,q,i,j,k;

void read_matrix(int a[50][50],int m,int n)
{
    for(i=0;i<m;i++)
    {
        for(j=0;j<n;j++)
        {
            scanf("%d",&a[i][j]);
        }
    }
}

void print_matrix(int a[50][50],int m,int n)
{
    for(i=0;i<m;i++)
    {
        for(j=0;j<n;j++)
        {
            printf("%d\t",a[i][j]);
        }
        printf("\n");
    }
}

void compute_matrix(int a[50][50], int b[50][50], int m,int q)
{
    for(i=0;i<m;i++)
    {
        for(j=0;j<q;j++)
        {
            c[i][j]=0;
            for(k=0;k<n;k++)
            {
                c[i][j]=c[i][j]+a[i][k]*b[k][j];
            }
        }
    }
}
```



```
void main()
{
    printf("Enter the order of Matrix A\n");
    scanf("%d%d",&m,&n);
    printf("Enter the order of Matrix B\n");
    scanf("%d%d",&p,&q);
    if(n!=p)
    {
        printf("Matrix Multiplication not possible\n");
    }
    else
    {
        printf("Enter the elements of matrix A\n");
        read_matrix(a,m,n);
        printf("Matrix A is displayed as\n");
        print_matrix(a,m,n);
        printf("Enter the elements of matrix B\n");
        read_matrix(b,p,q);
        printf("Matrix B is displayed as\n");
        print_matrix(b,p,q);
        printf("\nMatrix Multiplication\n");
        compute_matrix(a,b,m,q);
        printf("Resultant matrix \n");
        print_matrix(c,m,q);
    }
}
```

Develop a C Program to Implement structures to read, write and compute average- marks of the students, list the students scoring above average marks for a class of N students

```
#include<stdio.h>
#include<string.h>

struct student
{
    int rollno;
    char name[20];
    int marks;
    char grade[1];
};

void main()
{
    struct student s[20];
    char sname[20];
    int i,n;
    float sum=0,avg=0;
    printf("Enter the number of students\n");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("\nEnter student %d details\n",i+1);
        printf("\nEnter Roll Number , Name , Marks and Grade\n");
        scanf("%d%s%d %s",&s[i].rollno, s[i].name, &s[i].marks, s[i].grade);
        sum+=s[i].marks;
    }
    avg=sum/n;
    printf("\nAverage Marks= %f\n",avg);
    printf("\n Student Details are \n");
    printf("ROLL_NO.\t NAME\t MARKS\t GRADE\n");
    for(i=0;i<n;i++)
    {
        printf("%d\t\t%s\t%d\t%s\n",s[i].rollno,s[i].name,s[i].marks,s[i].grade);
    }
    printf("\nDetails of students who scored above Average Marks\n");
    printf("ROLL_NO.\t NAME\t MARKS\t GRADE\n");

    for(i=0;i<n;i++)
    {
        if(s[i].marks>=avg)

printf("%d\t\t%s\t%d\t%s\n",s[i].rollno,s[i].name,s[i].marks,s[i].grade);
    }
}
```

Develop a C program to copy a text file to another, read both the input file name and target file name

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char inputFileName[100];
```

```
    char outputFileName[100];
```

```
    char ch;
```

```
    FILE *inputFile, *outputFile;
```

```
    printf("Enter the input file name: ");
```

```
    scanf("%s", inputFileName);
```

```
    inputFile = fopen(inputFileName, "r");
```

```
    if (inputFile == NULL)
```

```
    {
```

```
        printf("Error: Cannot open input file.\n");
```

```
        return 1;
```

```
    }
```

```
    printf("Enter the output file name: ");
```

```
    scanf("%s", outputFileName);
```

```
    outputFile = fopen(outputFileName, "w");
```

```
    if (outputFile == NULL)
```

```
    {
```

```
        printf("Error: Cannot open output file.\n");
```

```
        fclose(inputFile);
```

```
        return 1;
```

```
    }
```

```
    while ((ch = fgetc(inputFile)) != EOF)
```

```
    {
```

```
        fputc(ch, outputFile);
```

```
    }
```

```
    fclose(inputFile);
```

```
    fclose(outputFile);
```

```
    printf("File copied successfully.\n");
```

```
    return 0;
```

```
}
```